

SIMATS ENGINEERING

CO4 - ASSESSMENT TOOL 2

Industry Problem Solving Task

COURSE NAME: Deep Learning

COURSE CODE: MLA0408

COURSE OUTCOME COVERED

CO 1: Students will be able to understand and apply probabilistic graphical models including Bayesian Networks, Markov Random Fields, Hidden Markov Models, and Conditional Random Fields. They will learn how to represent complex probabilistic relationships among variables and perform inference and learning in graphical models. Students will be able to use these models for reasoning under uncertainty and solving real-world decision-making problems. (BL-6)

Assessment Tool Weightage: 40%

QUESTIONS:

Total Marks:25

Q1. Transfer Learning

A healthcare startup aims to develop an AI-based system for detecting skin cancer from dermoscopic images. Since only a limited number of labeled images are available, design a transfer learning strategy using a pre-trained CNN model. Justify your choice of model, identify which layers should be frozen or fine-tuned, and explain how your design improves classification performance.

Q2. DenseNet and PixelNet

A smart city project requires an automated road-scene analysis system that accurately identifies roads, pedestrians, vehicles, and traffic signs at the pixel level. Design a deep learning architecture using DenseNet and PixelNet to solve this problem. Explain how the proposed architecture improves feature reuse, segmentation accuracy, and computational efficiency.

Q3. Bidirectional RNN and Encoder–Decoder

A multinational customer service company wants to build a multilingual chatbot capable of translating customer queries while preserving contextual meaning. Design an Encoder–Decoder sequence-to-sequence model using Bidirectional RNNs. Illustrate the architecture and justify how bidirectional processing improves translation accuracy.

Q4. BPTT and LSTM

A financial analytics company is developing a model to forecast stock prices using historical market data. Create an LSTM-based prediction system and explain how Backpropagation Through Time (BPTT) is used to train the network. Justify why LSTM is more suitable than a traditional RNN for this application.

Q5. Integrated Deep Learning Solution

A media streaming company plans to automatically generate captions for uploaded videos to improve accessibility. Design an integrated deep learning solution that combines Transfer Learning for visual feature extraction with an LSTM-based Encoder–Decoder architecture for

caption generation. Explain the role of each component and justify how the complete system produces accurate captions.

Q1. Transfer Learning – Skin Cancer Detection

Problem

A healthcare startup has only a limited number of labeled dermoscopic images. Training a CNN from scratch would require a large dataset and may lead to overfitting. Therefore, transfer learning using a pre-trained CNN is suitable.

Proposed Model: ResNet50

I would use ResNet50 pre-trained on ImageNet because:

It has strong image-feature extraction capability.

Residual connections help train deep networks.

It can learn edges, textures, shapes, and higher-level visual patterns.

Pre-trained features reduce training time and data requirements.

Architecture

Dermoscopic Image



Resize + Normalization



Data Augmentation



Pre-trained ResNet50



Early + Middle CNN Layers



Fine-tuned Deep Layers



Global Average Pooling



Fully Connected Layer



Sigmoid / Softmax



Benign / Malignant

Freezing and Fine-Tuning

Initially:

ResNet50 Layers

Early Layers → FROZEN



Middle Layers → FROZEN
↓
Last 20–30 Layers → FINE-TUNED
↓
New Classification Head → TRAINED

The early layers learn general features such as edges and textures, so they do not need to be retrained. The deeper layers learn task-specific features and should be fine-tuned using the skin-cancer dataset.

Training Strategy

- Load ResNet50 with ImageNet weights.
- Replace the original classification layer.
- Add Global Average Pooling.
- Add Dropout to reduce overfitting.
- Add a binary classification layer.
- Freeze most ResNet layers.
- Train the new classification head.
- Unfreeze the final ResNet layers.
- Fine-tune using a small learning rate.

Data Augmentation

Rotation
Flipping
Zooming
Cropping
Brightness Adjustment
↓
More Training Variations

Advantages

- Requires fewer labeled images.
- Reduces training time.
- Reduces overfitting.
- Improves feature extraction.
- Fine-tuning adapts ImageNet features to dermoscopic images.

Conclusion

Transfer learning with ResNet50 provides an efficient solution for skin-cancer classification when labeled medical data is limited. Freezing early layers and fine-tuning deeper layers allows the model to retain general visual knowledge while learning cancer-specific features.

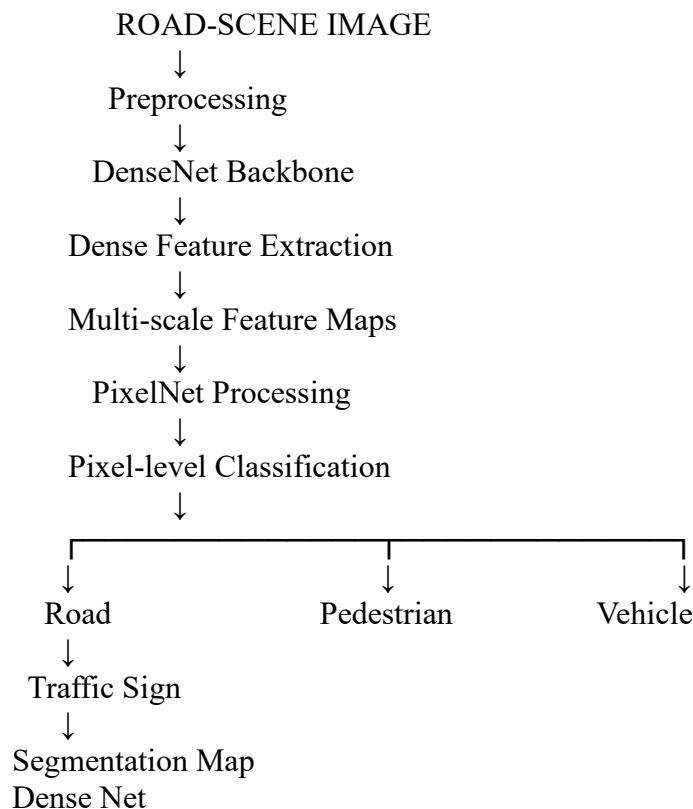
Q2. DenseNet and PixelNet – Road Scene Analysis

Problem

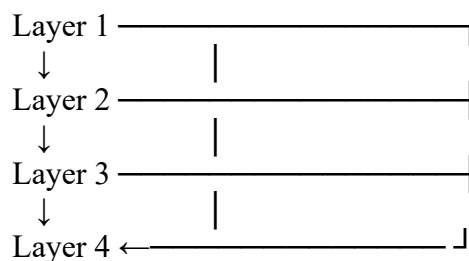
A smart-city system needs to identify roads, pedestrians, vehicles, and traffic signs at pixel level. Therefore, the task is semantic segmentation, where every pixel receives a class label.

Proposed Architecture

A DenseNet-based feature extractor can be combined with a PixelNet-style pixel-level prediction framework.



Dense Net connects each layer to subsequent layers.



Each layer receives feature maps from previous layers.

Feature Reuse

This provides:

Feature reuse

Better gradient propagation
Reduced redundant learning
Strong feature representation
PixelNet Component

PixelNet performs prediction at the pixel level.

For each pixel:

$P(y_i|x_i)$

is calculated to determine the class of that pixel.

For example:

Input Image

R R R R P P V V
R R R P P P V V
R R R R V V V V
S S R R V V T T

R = Road

P = Pedestrian

V = Vehicle

S = Sign

T = Traffic Sign

Improving Segmentation Accuracy

DenseNet provides rich features while PixelNet performs pixel-level classification.

The combination helps with:

Boundary detection

Small objects

Traffic signs

Pedestrians

Vehicle recognition

Road-region segmentation

Computational Efficiency

DenseNet promotes feature reuse instead of learning completely new features at every layer. This can reduce unnecessary parameters and improve parameter efficiency.

Conclusion

A DenseNet + PixelNet architecture provides strong feature reuse and pixel-level classification. It is suitable for smart-city applications such as autonomous driving, traffic monitoring, and intelligent transportation systems.

Q3. Bidirectional RNN and Encoder–Decoder Problem

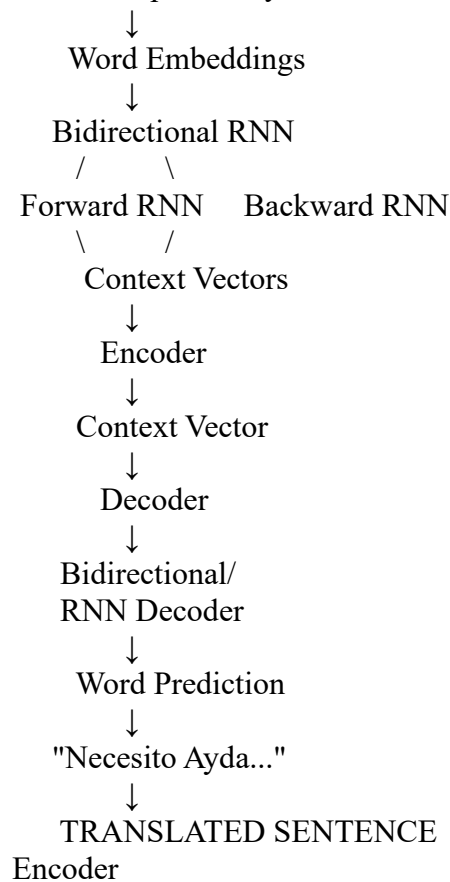
A multinational company requires a multilingual chatbot that can translate customer queries while preserving context.

An Encoder–Decoder sequence-to-sequence architecture with Bidirectional RNNs is suitable.

Architecture

SOURCE SENTENCE

"I need help with my order"



The encoder converts the input sentence into a representation.

For a sentence:

I → need → help → with → my → order

the Bidirectional RNN processes it in two directions.

Forward Direction

I → need → help → with → my → order

Backward Direction

order → my → with → help → need → I

The final representation contains information from both directions.

Bidirectional RNN

The hidden state can be represented as:

$ht=[hT;ht]$

where:

ht = forward hidden state

ht = backward hidden state

Decoder

The decoder receives the encoded representation and generates the translated sentence one word at a time.

Context Vector



Decoder



Word 1



Word 2



Word 3



...



End of Sentence

Why Bidirectional Processing?

Consider:

"I went to the bank to deposit money."

The meaning of "bank" depends on surrounding words.

A normal RNN primarily processes previous context, whereas a Bidirectional RNN can use information from both sides of each input position during encoding.

This improves:

- Context understanding
- Word disambiguation
- Translation accuracy
- Sentence representation
- Multilingual understanding

Conclusion

The Bidirectional RNN encoder captures contextual information from both directions, while the decoder converts the learned representation into the target language. This makes the system effective for contextual multilingual translation.

Q4. BPTT and LSTM – Stock Price Prediction Problem

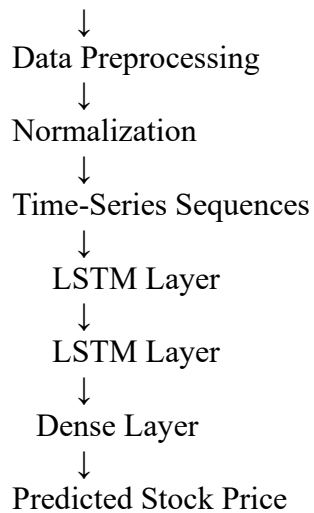
Stock prices depend on historical information such as:

Previous prices
Trading volume
Market trends
Moving averages
Historical returns
Other time-dependent indicators

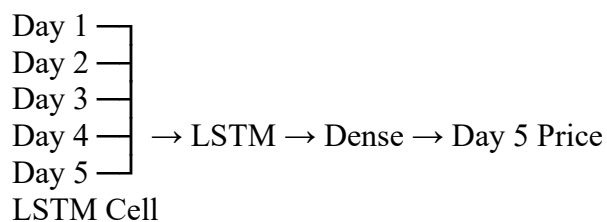
Therefore, stock prediction is a sequence-learning problem.

Proposed LSTM Architecture

Historical Market Data



For example:



An LSTM contains three major gates:

Forget Gate

Determines what information should be removed.

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f)$$

Input Gate

Determines what new information should be stored.

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i)$$

Output Gate

Determines what information should be output.

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o)$$

The cell state is updated using:

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t$$

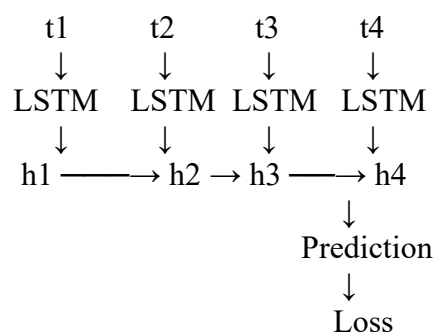
and the hidden state is:

$$h_t = o_t * \tanh(C_t)$$

BPTT – Backpropagation Through Time

BPTT is used to train recurrent networks.

The LSTM is conceptually unfolded across time:



The error is propagated backward through all time steps.

Prediction



Calculate Loss



Backward Through t4



Backward Through t3



Backward Through t2



Backward Through t1



Update Weights

Why LSTM Instead of Traditional RNN?

Traditional RNNs can suffer from:

Vanishing gradients

When sequences become long, information from earlier time steps may be lost.

LSTM uses gates and a cell-state mechanism to preserve important information for longer periods.

Therefore:

Traditional RNN	LSTM
Simple architecture	More complex architecture
Difficult with long-term dependencies	Handles long-term dependencies
More vulnerable to vanishing gradients	Reduces vanishing-gradient problem
Limited memory	Better long-term memory
Less suitable for long financial sequences	Better suited for time-series forecasting

Conclusion

LSTM is suitable for stock-price forecasting because financial data is sequential and may contain long-term temporal relationships. BPTT allows the network to learn from errors across multiple time steps, while LSTM gates help preserve important historical information.

Q5. Integrated Deep Learning Solution – Video Caption Generation Problem

A media company wants to automatically generate captions for uploaded videos to improve accessibility.

The system must understand:

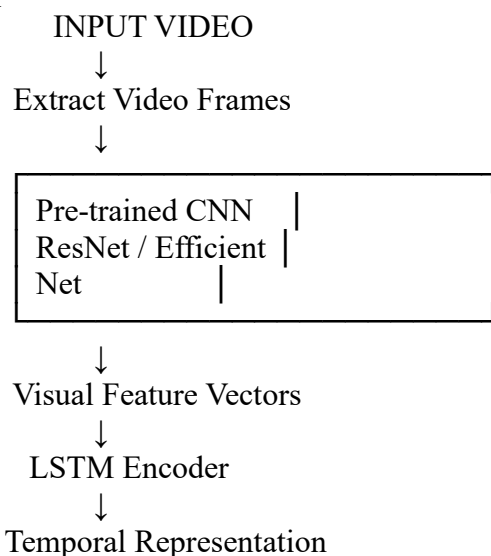
What is happening visually?

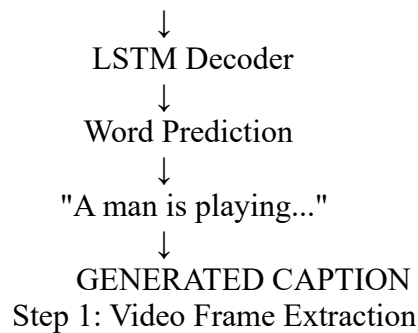
What is the temporal sequence of events?

How should those events be converted into natural language?

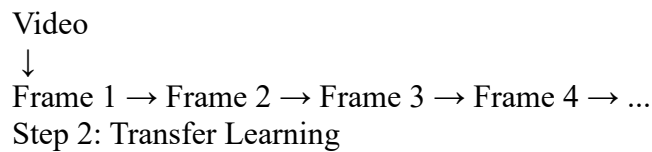
A combination of Transfer Learning + LSTM Encoder–Decoder can solve this problem.

Proposed Architecture



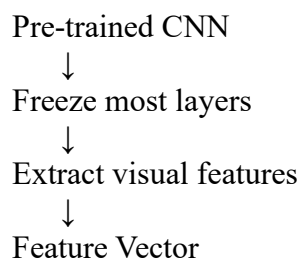


A video is divided into individual frames.



A pre-trained CNN such as ResNet50 is used to extract visual features.

Instead of training the CNN from scratch:



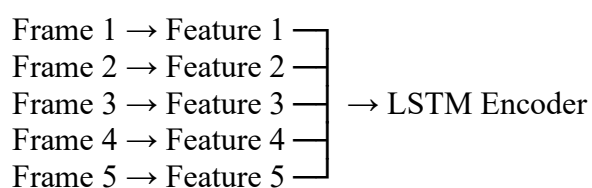
The CNN can recognize:

Objects
Shapes
Textures
People
Animals
Backgrounds

This is useful because large video-caption datasets may not always be available.

Step 3: LSTM Encoder

The extracted frame features are provided to the LSTM.



The LSTM learns the temporal relationship between frames.

For example:

Frame 1: Person

Frame 2: Person + Ball

Frame 3: Person + Ball

Frame 4: Person kicks Ball

The LSTM learns that these frames represent a sequence of actions.

Step 4: LSTM Decoder

The decoder converts the learned representation into words.

Visual Context

↓
<START>
↓
A
↓
man
↓
is
↓
kicking
↓
a
↓
ball
↓
<END>

The final output could be:

"A man is kicking a ball."

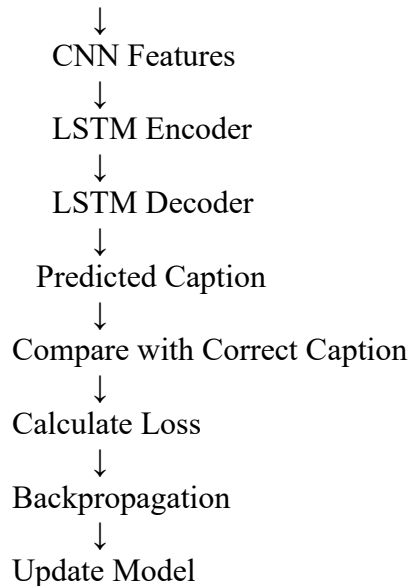
Role of Each Component

Component	Role
Video Input data	
Frame extraction	Converts video into images
Pre-trained CNN	Extracts visual features
Transfer learning	Reduces training requirements
LSTM Encoder	Learns temporal information
LSTM Decoder	Generates language
Word embeddings	Represent words numerically
Softmax	Predicts the next word

Training Process

During training, the model receives:

Video + Correct Caption



A categorical cross-entropy loss can be used:

$$L = -\sum_t y_t \log(y_t^{\wedge})$$

where:

y_t = actual next word

$y_t^{\wedge}$ = predicted probability

Why This Integrated Approach Works

Transfer Learning

Provides powerful visual representations without requiring the CNN to be trained from scratch.

LSTM Encoder

Understands the temporal relationship between video frames.

LSTM Decoder

Converts visual-temporal information into meaningful natural-language sentences.

Complete Process

Video → CNN → Features → LSTM Encoder → LSTM Decoder → Caption

Example

Input Video: A child riding a bicycle in a park.

Frames
↓
CNN
↓
Child + Bicycle + Park
↓
LSTM Encoder
↓
Temporal Context
↓
LSTM Decoder
↓
"A child is riding a bicycle in a park."

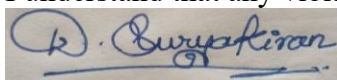
Conclusion

The combination of transfer learning and an LSTM Encoder–Decoder provides an effective solution for automatic video caption generation. The pre-trained CNN extracts rich visual information, while the LSTM encoder learns temporal relationships between frames and the decoder generates a meaningful caption. This approach can improve accessibility for people with hearing difficulties and can also be used for video search, content indexing, and automated media description.

Code of Conduct:

I D. Suryakiran/192325057(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts

		effectively to solve the problem			
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis